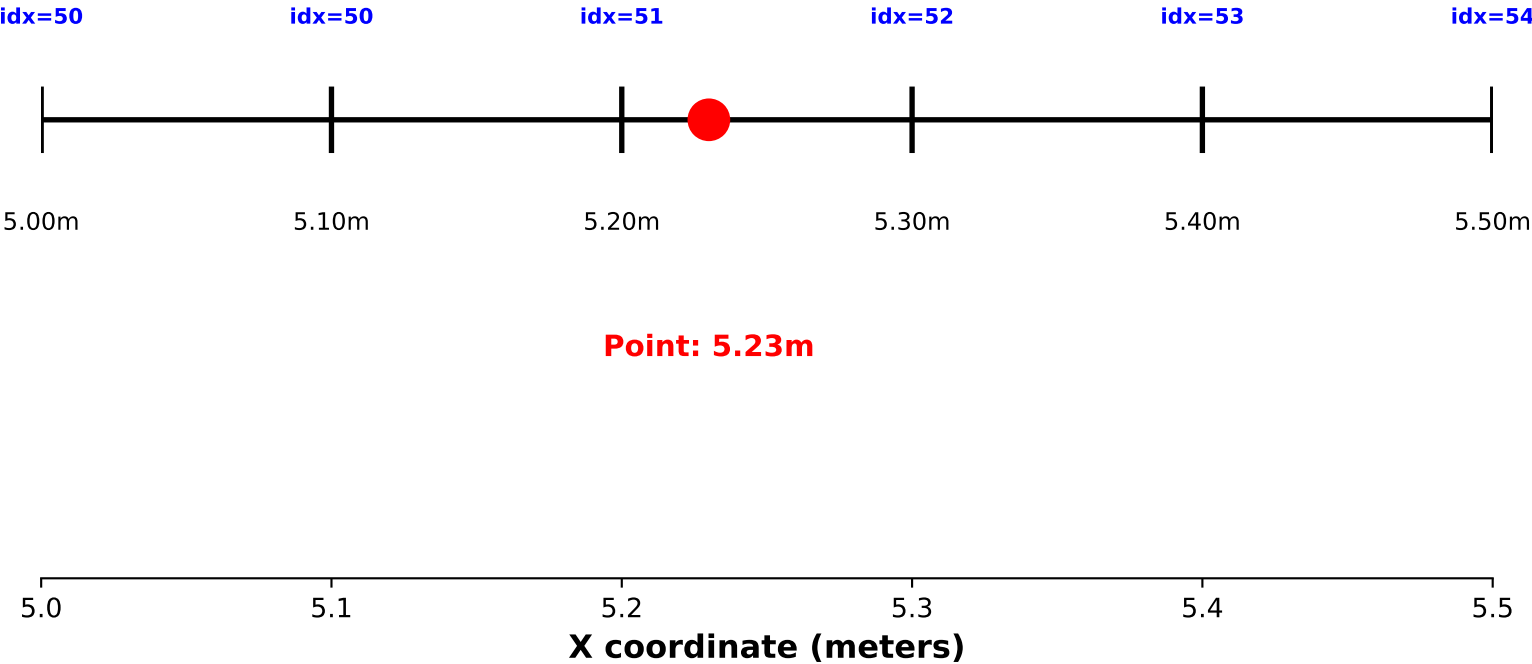


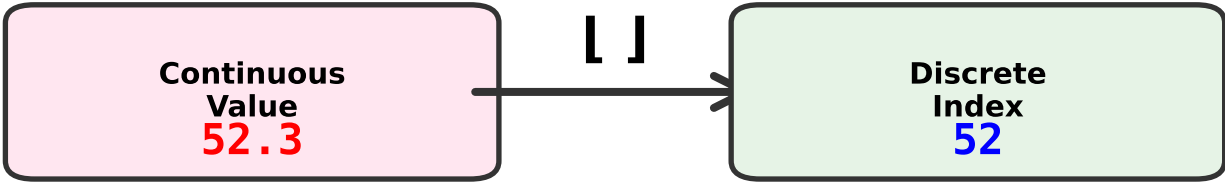
Floor Division in Voxelization: Step-by-Step Mathematical Process

Step 1: Division (Point Position / Voxel Size)

5.23 ÷ 0.1 = 52.3

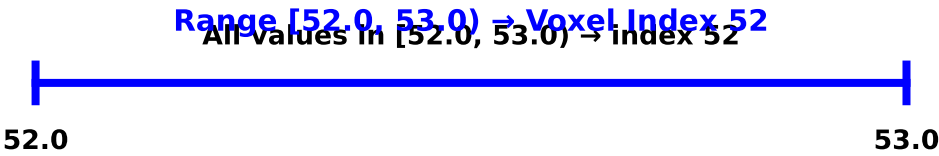


Step 2: Floor Function ([52.3])



Floor function $\lfloor x \rfloor$ rounds DOWN to nearest integer:

$\lfloor 52.3 \rfloor = 52$ (not 52.3)
 $\lfloor 52.7 \rfloor = 52$ (not 53)
 $\lfloor 52.0 \rfloor = 52$
 $\lfloor 52.9 \rfloor = 52$



Step 3: Apply to All 3 Dimensions (X, Y, Z)

Point Position:

P = (5.23, 3.67, 1.42) meters

Voxel Size:

$\sigma = 0.10$ meters

X dimension:

5.23 ÷ 0.10 = 52.3 $\lfloor 52.3 \rfloor = 52$

Y dimension:

3.67 ÷ 0.10 = 36.7 $\lfloor 36.7 \rfloor = 36$

Z dimension:

1.42 ÷ 0.10 = 14.2 $\lfloor 14.2 \rfloor = 14$

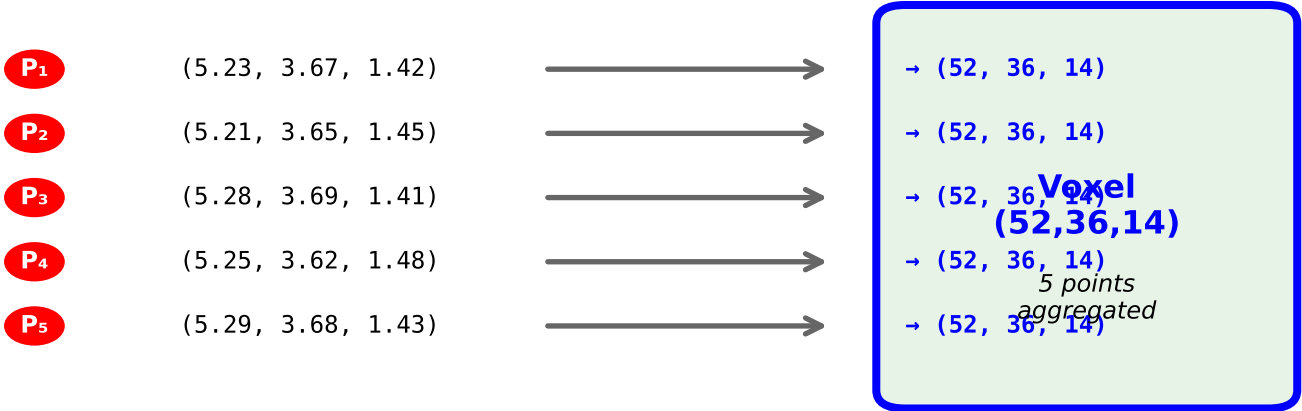
Voxel Index:

i = (52, 36, 14)

☐ Voxel (52, 36, 14) contains all points in region:

X: [5.20, 5.30) m • Y: [3.60, 3.70) m • Z: [1.40, 1.50) m

Result: Many-to-One Generalization



Key Insight: Generalization via Floor Division

- Infinite continuous positions → Finite discrete indices
- All points in same 10cm³ cube → Same voxel index
- Reduces complexity but loses fine-grained position info